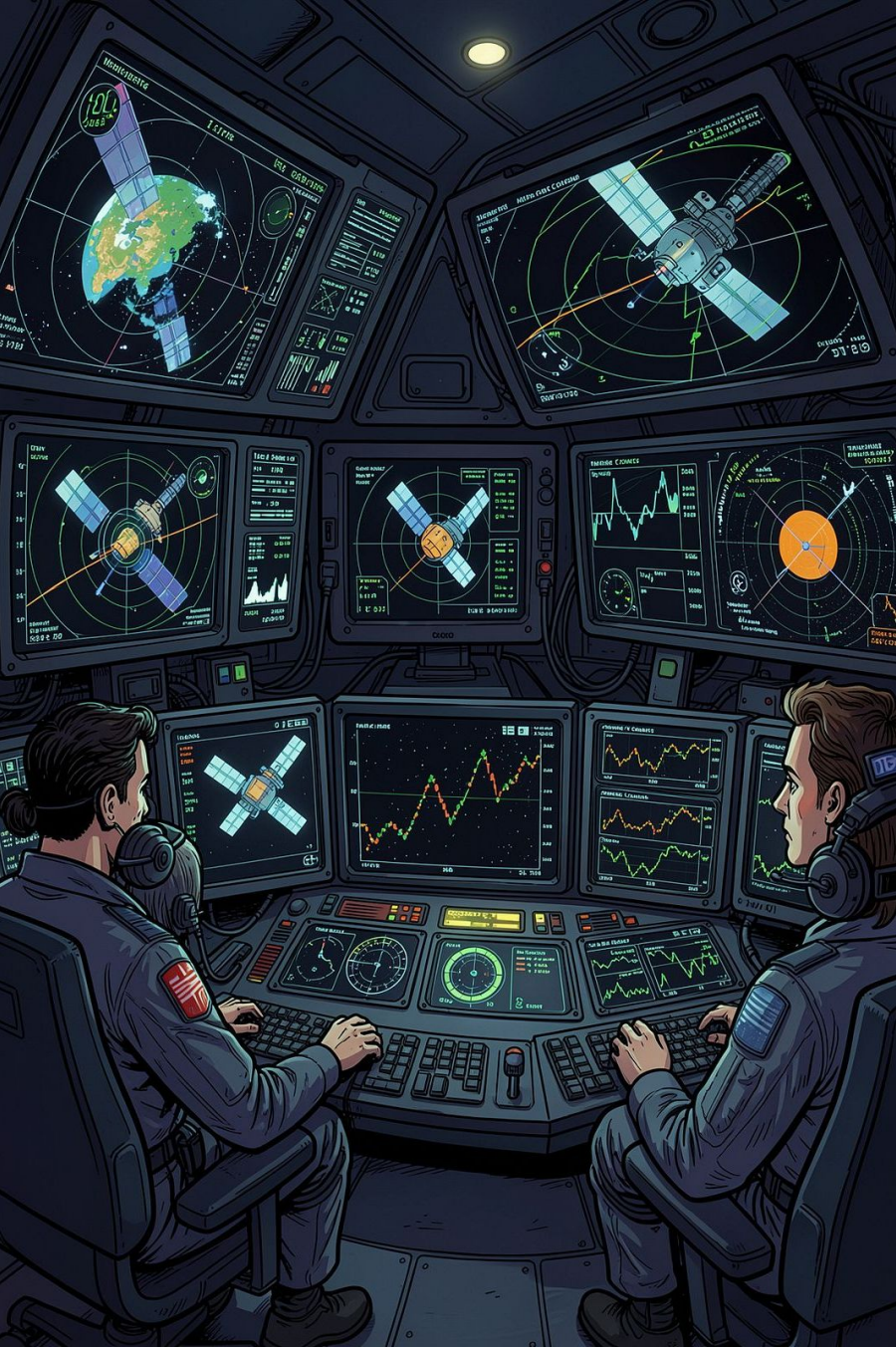




IBM BOB AI HACKATHON · TEAM CODECRAFTERS

AssetSentinel

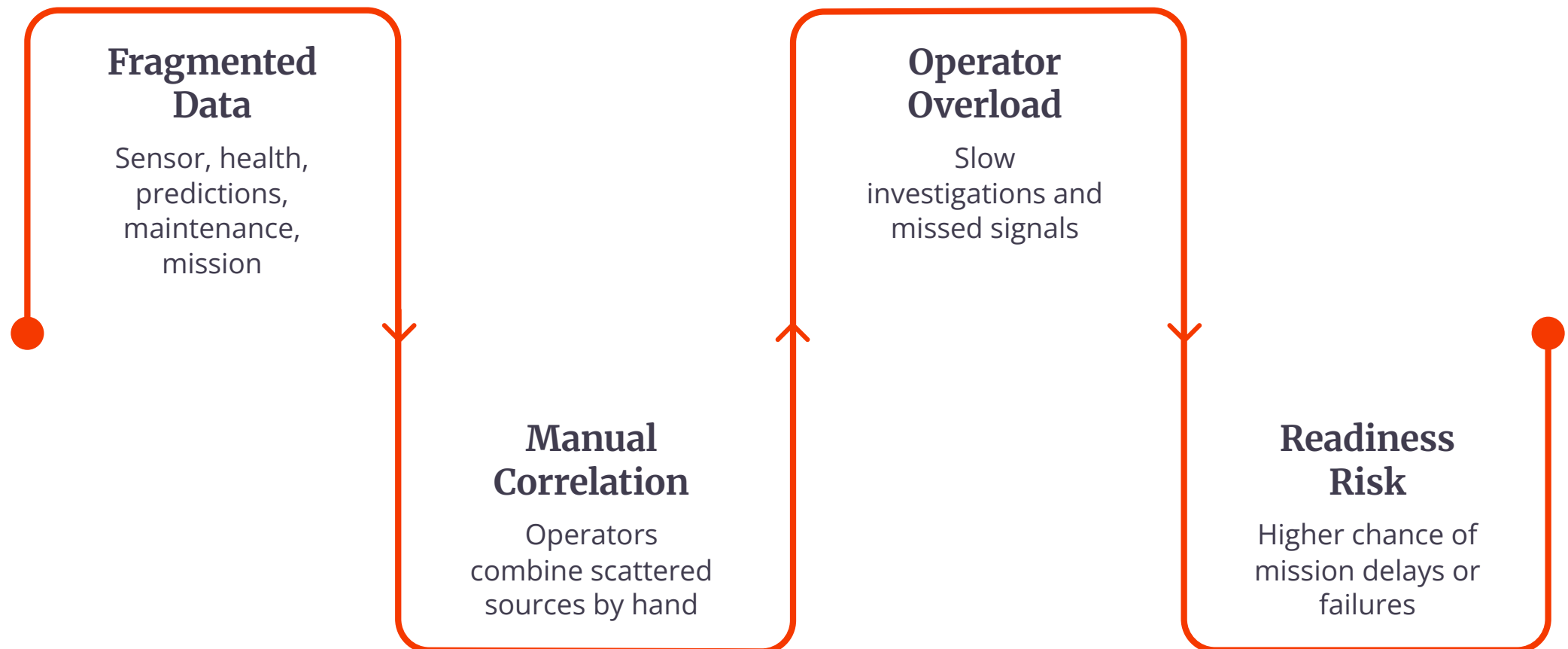
Mission Readiness & Predictive Maintenance Copilot

"Turning asset telemetry into mission-ready intelligence."

Made with GAMMA

The Mission Readiness Problem

Operators must determine whether assets are ready for missions — but critical information is scattered across sensor telemetry, component health, failure predictions, anomalies, maintenance history, and mission requirements. Manually correlating these sources creates **operator overload** and **mission readiness risk**.



Operators need **one place** to understand what is wrong, why it matters, and what needs attention first.

AssetSentinel — One Copilot for Mission Readiness

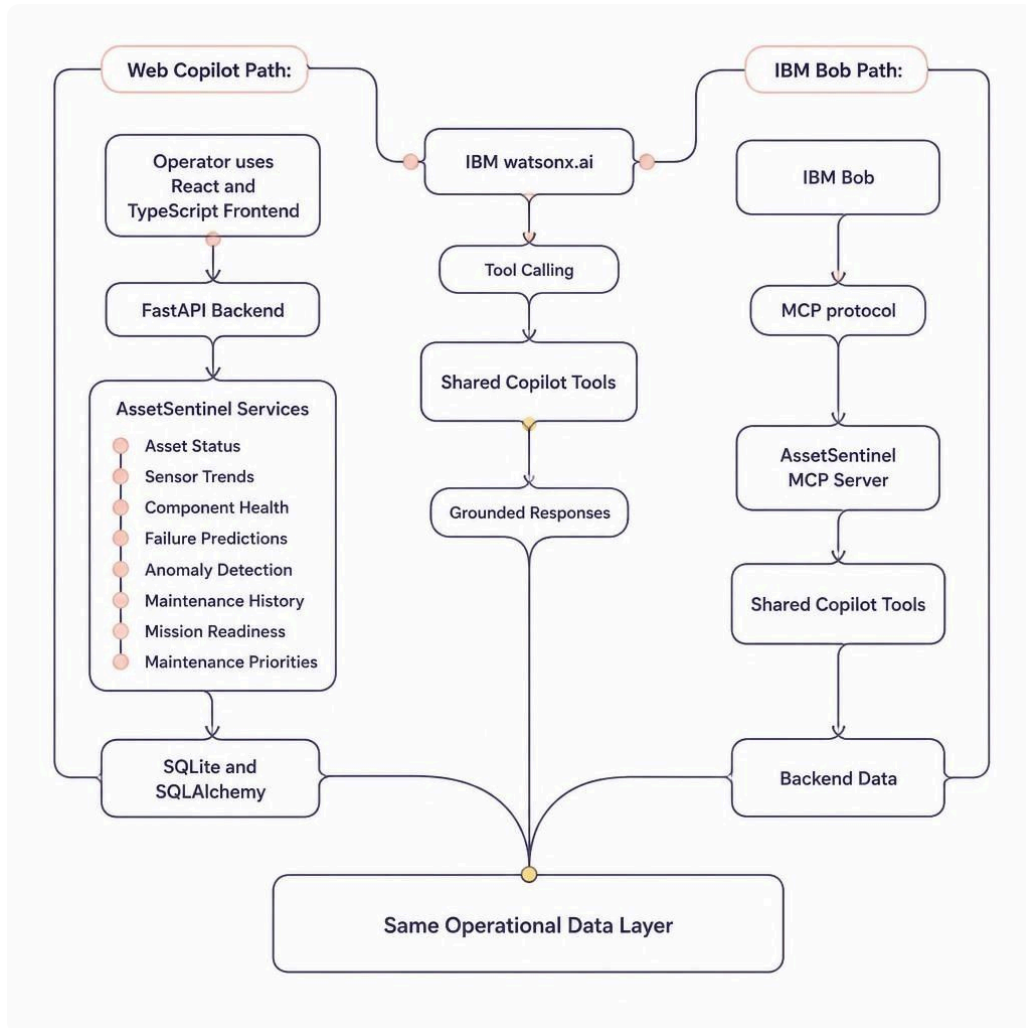
AssetSentinel combines operational data, machine learning, evidence, readiness logic, and conversational AI into a single mission-readiness and predictive-maintenance copilot.



"Ask questions in natural language. Get answers grounded in actual AssetSentinel data."

"What is the status of AS-1047?" · "Why is AS-1047 not ready?" · "Which component has the highest failure risk?" · "What anomalies were detected?" · "Which assets need immediate attention?"

How AssetSentinel Works



Two AI Paths, One Data Layer

IBM watsonx.ai powers the web conversational copilot through tool calling and grounded responses.

IBM Bob connects through MCP — discovering and invoking AssetSentinel's operational tools directly.

Both paths share the same backend services and live asset data.

From Question to Mission Insight

Operator Question

"Why is AS-1047 not ready?"

Also try: *"What anomalies were detected on AS-1047?"*

Responses are **grounded in backend data** — not hardcoded answers.

How AssetSentinel Responds

01

Retrieves asset status

02

Checks component health

03

Checks failure predictions

04

Checks anomalies & sensor trends

05

Reviews maintenance evidence

06

Evaluates readiness evidence

07

watsonx.ai explains the result

IBM Technology Integration

IBM watsonx.ai — Web Copilot

- Real watsonx.ai integration for conversational AI
- Natural-language questions with multi-turn context
- Tool calling to live AssetSentinel backend
- Grounded responses — not generic AI text

User → watsonx.ai → Tools → Live Data → Explanation

IBM Bob + MCP — Agent Integration

- Real IBM Bob MCP integration
- AssetSentinel exposes a real MCP server
- IBM Bob discovers and invokes operational tools
- Tools are read-only and backed by project data

IBM Bob → MCP Server → AssetSentinel Tools → Backend Data

❏ **IBM watsonx.ai powers the web copilot, while IBM Bob is integrated through MCP.**

MCP Tools Exposed by AssetSentinel

get_asset_status

get_anomalies

get_fleet_readiness

get_failure_predictions

get_mission_readiness

get_maintenance_priorities

IMPACT

Why AssetSentinel Matters



Faster Investigation

Ask questions instead of navigating multiple data sources.



Earlier Risk Detection

Random Forest failure prediction and Isolation Forest anomaly detection identify potential problems.



Evidence-Based Readiness

Readiness decisions backed by asset, component, sensor, prediction, anomaly, and maintenance evidence.



Prioritized Maintenance

Identifies assets and maintenance priorities requiring attention.



Explainable AI

watsonx.ai explains evidence — not generic responses.

Current Prototype Capabilities

PREDICTIVE FAILURE RISK

ANOMALY DETECTION

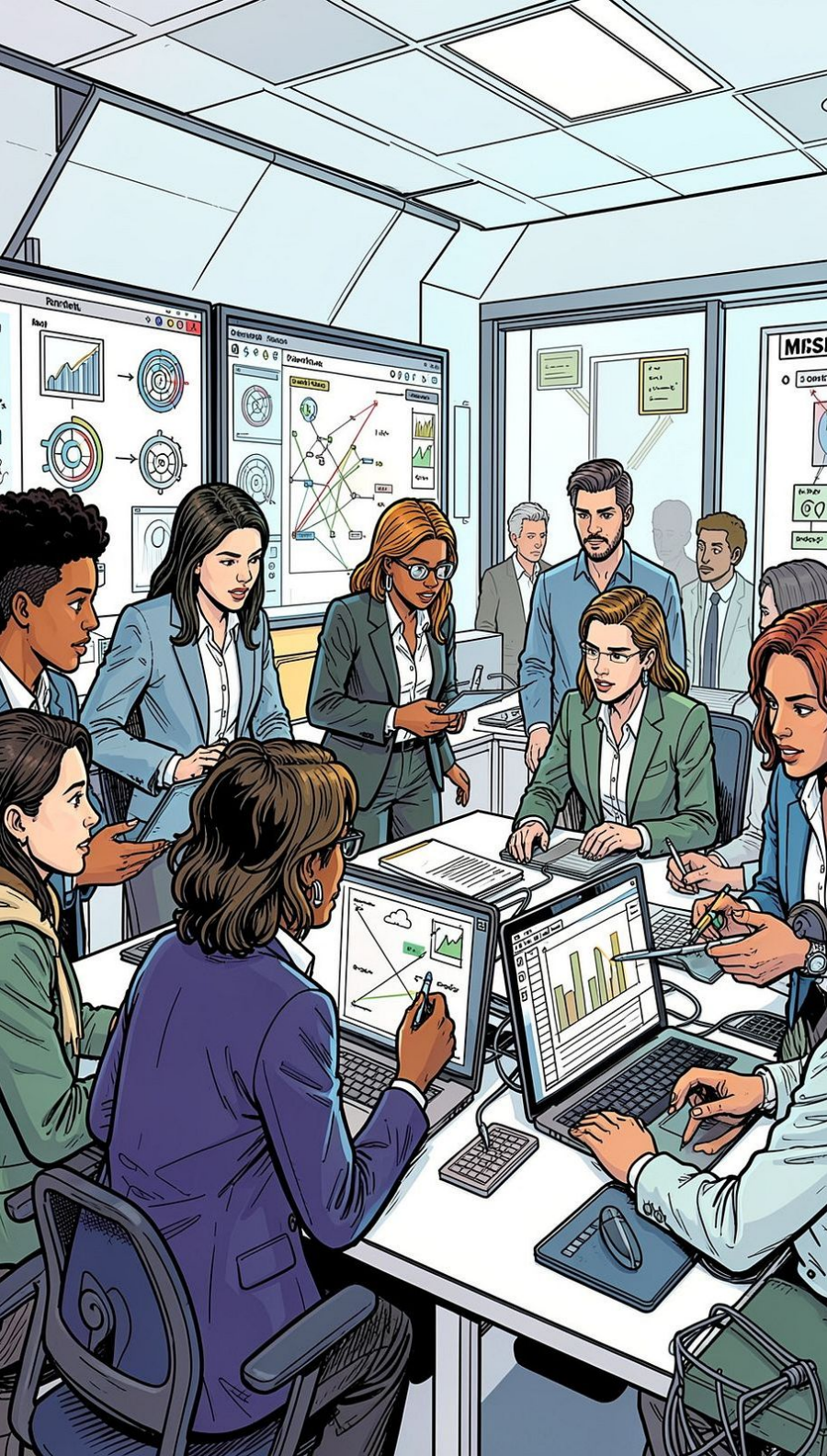
COMPONENT HEALTH

MISSION READINESS

MAINTENANCE PRIORITIES

CONVERSATIONAL AI

IBM BOB MCP INTEGRATION



TEAM CODECRAFTERS

Team Codecrafters

Tisha Soni

Role: READINESS + MAINTENANCE ENGINE + INTEGRATION

Selin Parmar

Role: ML + FAILURE PREDICTION + COPILOT

Dhruv Sutariya

Role: BACKEND + DATABASE + SECURITY

Mukt Patel

Role: FRONTEND

AssetSentinel — *Mission readiness through predictive intelligence.*